

Looped_Transformer

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1 Looped Transformer

1.1 理论

Looped Transformer 所有层共享相同的权重 θ ，每层的输入是前一层输出与初始输入的和：

$$\begin{aligned} h_0 &= \text{input} \\ h_l &= \text{TransformerBlock}(h_{l-1} + h_0 | \theta) \quad \text{for } l = 1, 2, \dots, L \end{aligned}$$

1.1.1 截断损失函数

$$\text{Loss} = \min_{\theta} \mathbb{E}_P \left[\frac{1}{b - b_0} \sum_{t=b_0}^b \frac{1}{k+1} \sum_{i=0}^k (Y_t(P^i | \theta) - f(x_{i+1}))^2 \right]$$

其中 - $\min_{\theta}$ ：这是我们在寻找一组最优的物理定律（模型参数 θ ）。 - $\mathbb{E}_P$ ：对所有生成的 Prompt 分布求期望。在统计物理中，这相当于求“系综平均”，即我们在无数种不同的初始条件（数据集）下测试这个定律是否普适。 - $\frac{1}{b-b_0} \sum_{t=b_0}^b$ ：这是时间平均。 b 是总迭代次数， T 是窗口大小， $b_0 = \max(b - T, 0)$ 。我们只关心系统演化到“稳态”附近的这段时间内（也就是 b_0 到 b 的表现），前面瞬态的误差我们不计较。 - $\frac{1}{k+1} \sum_{i=0}^k$ ：这是空间/序列平均。也就是在给定不同长度的上下文 P^i 时，模型预测下一个点的值与真实函数 $f(x_{i+1})$ 之间的均方误差。 - $Y_t(P^i | \theta)$ ：在时间 t 、给定历史信息 P^i 和系统参数 θ 时，系统当前的状态输出。

1.1.2 下游任务

- 数据生成方式：这篇论文解决的是纯粹的数学数据拟合问题。数据集是在训练循环中即时生成的浮点数向量。
- 以最基础的 **线性回归（Linear Regression）** 为例：
 - 维度设定：问题维度 $d = 20$ ，上下文样本数 $k = 40$ 。
 - 生成逻辑：每次迭代前，随机采样一个真实的参数 $w \sim \mathcal{N}(0, I_d)$ 。
 - 采样输入：生成 $k + 1$ 个输入样本（包含测试样本） $x_i \sim \mathcal{N}(0, I_d)$ 。

- 计算标签: $y_i = w^T x_i$ 。
- 拼接成 Prompt: 最终喂给模型的序列 P 是一个交替的序列 $(x_1, y_1, x_2, y_2, \dots, x_k, y_k, x_{test}, y_{dummy})$ 。
- 线性回归可以进一步改成非线性回归 (比如 ReLU 与 Linear 叠加), 或者更复杂的物理系统。

把 Prompt 映射到高维 Embedding 空间 双通道映射与“拉链式”交织

由于 x 和 y 代表不同的物理量 (一个是空间坐标/特征, 一个是目标观测值), 我们需要赋予它们独立的投影空间:

1. 定义两个独立的线性层:

```
read_in_x = nn.Linear(20, 256)
read_in_y = nn.Linear(1, 256)
```

2. 把前 k 个 x_i 丢给 `read_in_x`, 变成张量 X_{emb} , 形状为 $[batch, k, 256]$ 。
3. 把前 k 个 y_i 丢给 `read_in_y`, 变成张量 Y_{emb} , 形状为 $[batch, k, 256]$ 。
4. 像拉拉链一样, 在 `seq_len` 维度上把 X_{emb} 、 Y_{emb} 交织 (Interleave) 起来, 拼接成最终长度为 $2k$ 的提示向量 P 。

在工程实现中, 我们在 x_{test} 后填充了一个 y_{dummy} (如 0), 使实际拼接长度变为 $2k+2$ (偶数)。因此, 真正对 x_{test} 的预测结果位于输出序列的倒数第二个位置, 即索引 $[-2]$ 处。

1.2 代码

```
[53]: import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import time
```

1.2.1 绝对位置编码 (APE, Absolute Position Encoding)

```
[54]: class APE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096, b=10000):
        super().__init__()
        theta_i = 1 / (b ** (torch.arange(0, d_model, 2).float() / d_model)) # ↵
        ↵ [d_model/2]
        m = torch.arange(max_seq_len).float() # [max_seq_len]
        m_theta_i = torch.outer(m, theta_i) # [max_seq_len, d_model/2]
```

```

# APE 核心：直接生成一个完整的 pe 矩阵，偶数维度放 sin，奇数维度放 cos
stacked=torch.stack((torch.sin(m_theta_i), torch.cos(m_theta_i)),  

↳dim=-1) # [max_seq_len, d_model/2, 2]
pe=stacked.flatten(1,2) # [max_seq_len, d_model]
self.register_buffer('pe', pe.unsqueeze(0)) # shape: [1, max_seq_len,  

↳d_model]

def forward(self, x):
    # x shape: [batch_size, seq_len, d_model]
    return x + self.pe[:, :x.shape[1], :]

```

Learned APE

```

[55]: class LearnedAPE(nn.Module):
    def __init__(self, d_model, max_seq_len=4096):
        super().__init__()
        self.pe = nn.Embedding(max_seq_len, d_model)

    def forward(self, x):
        # x shape: [batch_size, seq_len, d_model]
        seq_len = x.shape[1]
        positions = torch.arange(seq_len, device=x.device)
        return x + self.pe(positions).unsqueeze(0)

```

ALiBi (Attention with Linear Biases) ALiBi 通过更改内积的方式引入线性位置偏置，从而实现了位置编码的功能。

$$\text{Score}(q_i, k_j) = q_i \cdot k_j - m \cdot (i - j) \quad (i \geq j)$$

其中 m 是一个与头数 (H) 相关的斜率参数，硬编码为

$$m_h = 2^{-\frac{8h}{H}} \quad h = 0, 1, \dots, H - 1$$

```

[56]: class ALiBi(nn.Module):
    def __init__(self, num_heads, max_seq_len=4096):
        super().__init__()
        dist = torch.arange(max_seq_len) # [max_seq_len]
        dist_matrix = torch.abs(dist.view(-1, 1) - dist.view(1, -1)) #  

↳[max_seq_len, max_seq_len]

```

```

        bias = -(2**-(8*torch.arange(1,num_heads+1) / num_heads)).view(-1, 1,
↪1) * dist_matrix.unsqueeze(0) # [num_heads, max_seq_len, max_seq_len]
        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
↪tril (triangular lower) 提取下三角元素, 并把上三角元素置 0
        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
↪float('-inf')) # 下三角元素置 0, 上三角元素置 -inf
        self.register_buffer('fused_mask', bias + mask) # [num_heads,
↪max_seq_len, max_seq_len]

    def forward(self, seq_len):
        # score shape: [batch_size, num_heads, seq_len, seq_len]
        return self.fused_mask[:, :seq_len, :seq_len]

```

原来的 RoPE (Rotary Position Encoding)

```

[57]: class RoPE(nn.Module):
    def __init__(self, d_k, max_seq_len = 4096, b=10000):
        super().__init__()
        theta_i = 1/(b**(torch.arange(0,d_k,2).float()/d_k)) # \theta_i =
↪b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        m = torch.arange(max_seq_len).float() # m = [0,1,...,seq_len-1]
        m_theta_i = torch.outer(m, theta_i) # [seq_len, d_k/2]
        cos = torch.cos(torch.cat((m_theta_i, m_theta_i), dim=-1)) # [seq_len,
↪d_k]
        sin = torch.sin(torch.cat((m_theta_i, m_theta_i), dim=-1)) # [seq_len,
↪d_k]
        self.register_buffer('cos', cos[None, None, :, :]) # 好处: cos 和 sin
        # 不需要更新参数, 注册为 buffer 后会自动放到正确的设备上
        self.register_buffer('sin', sin[None, None, :, :]) # 扩充维度以适应后续
        # 计算

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
        seq_len = x.shape[2]
        d_2 = x.shape[-1] // 2
        cos = self.cos[:, :, :seq_len, :] # [1, 1, seq_len, d_k]

```

```

sin = self.sin[:, :, :seq_len, :]
x_first_half = x[..., :d_2]
x_second_half = x[..., d_2:]
x_flip = torch.cat((-x_second_half, x_first_half), dim=-1)
x_out = x * cos + x_flip * sin # 旋转位置编码
return x_out

```

自创：Multi-Scale Untied Positional Encoding (MS-UPE)

```

[58]: class MS_UPE(nn.Module):
    def __init__(self, num_heads, d_k, max_seq_len = 4096, b_0=10000,
        ↪ head_ratio=2):
        super().__init__()
        b=(b_0*(head_ratio**(torch.arange(0, num_heads))))).float() # [num_heads]
        # \theta_i = b^{-\frac{2i}{d}}, \quad i \in \{0,1,\dots,\frac{d}{2}-1\}
        theta_i = 1/(b.unsqueeze(1)**(torch.arange(0,d_k,2).float()/d_k).
        ↪ unsqueeze(0)) # [num_heads, d_k/2]
        m = torch.arange(max_seq_len).float() # m = [0,1,...,seq_len-1]
        m_theta_i = torch.einsum('m,hk->hmk', m, theta_i) # [num_heads,
        ↪ seq_len, d_k/2]
        pe = torch.cat((torch.cos(m_theta_i), torch.sin(m_theta_i)), dim=-1) #
        ↪ [num_heads, seq_len, d_k]
        self.register_buffer('pe', pe[None, :, :, :]) # 好处: pe 不需要更新参数,
        注册为 buffer 后会自动放到正确的设备上

    def forward(self, x): # 适用于 Q、K (V 不需要位置编码!)
        # x: [batch_size, num_heads, seq_len, d_k]
        seq_len = x.shape[2]
        pe = self.pe[:, :, :seq_len, :] # [1, num_heads, seq_len, d_k]
        x_out = x + pe
        return x_out

```

1.2.2 去掉 RoPE 的 Multi-Head Attention

注：APE 应该加在 ToyModel 上而不是 TransformerBlock 上，因为我们只需要最开头输入处编码位置，而不是叠加 num_blocks 次。

```
[59]: class MultiHeadAttention(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096,
        ↪pe_type='learned_ape', b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        assert d_model % num_heads == 0, "d_model must be divisible by
        ↪num_heads"

        # 维度属性
        self.num_heads = num_heads
        self.d_model = d_model
        self.d_k = d_model // num_heads

        # 权重矩阵
        self.W_Q = nn.Linear(d_model, d_model, bias=False)
        self.W_K = nn.Linear(d_model, d_model, bias=False)
        self.W_V = nn.Linear(d_model, d_model, bias=False)
        self.W_O = nn.Linear(d_model, d_model, bias=False)

        # 位置编码模块
        self.pe_type = [pe_type] if isinstance(pe_type, str) else pe_type
        self.qk_pe_modules = nn.ModuleList()
        self.sc_pe_modules = nn.ModuleList()
        for pe in self.pe_type:
            if pe == 'rope':
                pe_module = RoPE(self.d_k, max_seq_len, b=b_rope_or_upe)
                self.qk_pe_modules.append(pe_module) # RoPE 应用于 Q 和 K
            elif pe == 'ms_upe':
                pe_module = MS_UPE(num_heads, self.d_k, max_seq_len,
                ↪b_0=b_rope_or_upe, head_ratio=head_ratio_upe)
                self.qk_pe_modules.append(pe_module) # MS-UPE 应用于 Q 和 K
            elif pe == 'alibi':
                pe_module = ALiBi(num_heads, max_seq_len)
                self.sc_pe_modules.append(pe_module) # ALiBi 应用于 score 矩阵
            elif pe in ['ape', 'learned_ape']:
                pass # APE 和 Learned APE 在输入时直接加到 x 上, 不需要单独模块

        # 因果掩码
        tril = torch.tril(torch.ones(max_seq_len, max_seq_len)).bool() # torch.
        ↪tril (triangular lower) 提取下三角元素, 并把上三角元素置 0
```

```

        mask = torch.zeros(max_seq_len, max_seq_len).masked_fill(~tril,
↪float('-inf')) # 下三角元素置 0, 上三角元素置-inf
        self.register_buffer('mask', mask[None, None, :, :]) # [1, 1, seq_len,
↪seq_len]

    def forward(self, x): # x: [batch_size, seq_len, d_model]
        # 线性投影与分头
        batch_size, seq_len, _ = x.shape
        Q = self.W_Q(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # [batch_size, num_heads, seq_len, d_k]
        K = self.W_K(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # 交换 (1, 2) 是因为之后 softmax 时要进行矩阵乘法 (只乘后两维)
        V = self.W_V(x).view(batch_size, seq_len, self.num_heads, self.d_k).
↪transpose(1, 2) # 在 seq_len 维度上分配注意力
        # 根据选择的 PE 类型对 Q 和 K 进行位置编码
        if not self.training:
            # 缩放点积注意力与因果掩码 Causal Masking
            ## Query-Key 点积计算注意力得分
            scores = torch.matmul(Q, K.transpose(-2, -1)) / (self.d_k ** 0.5)
↪# [batch_size, num_heads, seq_len, seq_len]
            ## 因果掩码和 softmax 计算注意力权重
            attention = F.softmax(scores + self.mask[..., :seq_len, :seq_len],
↪dim=-1) # [batch_size, num_heads, seq_len, seq_len]
            self.captured_attention = attention # ** 新增 **: 捕获注意力矩阵, 用
hook 记录下来, 方便后续分析
            ## 乘以 Value 完成加权求和
            out = torch.matmul(attention, V) # [batch_size, num_heads,
↪seq_len, d_k]
        else:
            for pe_module in self.qk_pe_modules:
                Q = pe_module(Q)
                K = pe_module(K)
            if self.sc_pe_modules:
                fused_mask = self.sc_pe_modules[0](seq_len) # [num_heads,
↪seq_len, seq_len]

```

```

        out = F.scaled_dot_product_attention(Q, K, V,
↪attn_mask=fused_mask, is_causal=False) # [batch_size, num_heads, seq_len,
↪d_k]

        else:
            out = F.scaled_dot_product_attention(Q, K, V, is_causal=True)
↪# [batch_size, num_heads, seq_len, d_k]
            # 多头拼接并乘以输出矩阵 (必须先内存连续化 (contiguous) 再 view, 否则会报错)

            out = out.transpose(1, 2).contiguous().view(batch_size, seq_len, -1)
↪# [batch_size, seq_len, d_model]
            H = self.W_O(out) # [batch_size, seq_len, d_model]
            return H

```

1.2.3 SwiGLU 和单个 Transformer Block 架构不变

```

[60]: class SwiGLU(nn.Module):
    def __init__(self, d_model, d_hidden):
        super().__init__()
        self.W = nn.Linear(d_model, d_hidden, bias=False)
        self.V = nn.Linear(d_model, d_hidden, bias=False)
        self.W2 = nn.Linear(d_hidden, d_model, bias=False)
    def forward(self, x):
        x1 = F.silu(self.W(x))
        x2 = self.V(x)
        x_out = self.W2(x1 * x2)
        return x_out

```

```

[61]: class TransformerBlock(nn.Module):
    def __init__(self, num_heads, d_model, max_seq_len=4096, multiplier=4,
↪norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
↪b_rope_or_upe=10000, head_ratio_upe=2):
        super().__init__()
        if norm_type == 'rmsnorm':
            self.norm1 = nn.RMSNorm(d_model) # 第一次归一化
            self.norm2 = nn.RMSNorm(d_model) # 第二次归一化
        elif norm_type == 'layernorm':
            self.norm1 = nn.LayerNorm(d_model) # 第一次归一化

```



```

        self.norm2 = nn.LayerNorm(d_model) # 第二次归一化
        self.attention = MultiHeadAttention(num_heads, d_model,
↪max_seq_len=max_seq_len, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
↪head_ratio_upe=head_ratio_upe)
        if ffn_type == 'swiglu':
            self.ffn = SwiGLU(d_model, d_model * multiplier)
        elif ffn_type == 'gelu':
            self.ffn = nn.Sequential(
                nn.Linear(d_model, d_model * multiplier),
                nn.GELU(),
                nn.Linear(d_model * multiplier, d_model)
            )

    def forward(self, x):
        # 第一次归一化 Pre-Norm
        x_norm1 = self.norm1(x)
        # 全局信息交互 Multi-Head Attention (MHA)
        h = self.attention(x_norm1)
        # 第一次残差连接
        x1 = x + h
        # 第二次归一化
        x_norm2 = self.norm2(x1)
        # 前馈网络 Feed-Forward Network (FFN)
        x_out = self.ffn(x_norm2)
        # 第二次残差连接
        x2 = x1 + x_out
        return x2

```

1.2.4 保留 Attention Probe

```

[62]: class AttentionProbe:
    def __init__(self):
        self.captured_data = []
    def __call__(self, module, input, output):
        if getattr(module, 'captured_attention', None) is not None:
            attention = module.captured_attention.detach().cpu().numpy()
            self.captured_data.append(attention)

```

```

        else:
            self.captured_data.append(None) # 如果没有捕获到注意力矩阵, 记录一个
None 占位符
    def reset(self):
        self.captured_data = []

```

1.2.5 Toy Model: Looped Transformer

```

[63]: class ToyModel(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, max_seq_len=4096,
                  norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
    ↪ b_rope_or_upe=10000, head_ratio_upe=2,
                  loop=True):
        super().__init__()
        self.loop = loop
        if loop:
            self.transformer_block = TransformerBlock(num_heads, d_model,
    ↪ max_seq_len=max_seq_len, norm_type=norm_type, ffn_type=ffn_type,
    ↪ pe_type=pe_type, b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
    ↪ # 这里保证了各层权重始终相同
            self.probe = AttentionProbe()
            self.transformer_block.attention.register_forward_hook(self.probe)
            self.captured_attention = None
        else:
            self.transformer_block = nn.ModuleList([
                TransformerBlock(num_heads, d_model, max_seq_len=max_seq_len,
    ↪ norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type,
    ↪ b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
                for _ in range(num_blocks)
            ])
            self.probe = [AttentionProbe() for _ in range(num_blocks)]
            for block, probe in zip(self.transformer_block, self.probe):
                block.attention.register_forward_hook(probe)

        pe_type = self.transformer_block.attention.pe_type
        self.ape = nn.ModuleList()

```

```

for pe in pe_type:
    if pe == 'ape':
        self.ape.append(APE(d_model, max_seq_len=max_seq_len))
    elif pe == 'learned_ape':
        self.ape.append(LearnedAPE(d_model, max_seq_len=max_seq_len))
    elif pe in ['rope', 'ms_upe']:
        pass # RoPE 和 MS_UPE 在 MultiHeadAttention 内部处理位置编码，不
需要单独的模块在这里处理

self.num_blocks = num_blocks
self.num_heads = num_heads
self.d_model = d_model
self.max_seq_len = max_seq_len
self.pe_type = pe_type

def forward(self, x_0, num_eff=15, current_blocks=None):
    # num_eff 是需要加入误差计算的有效层数，也即理论公式中的  $T$ 
    # x 的形状为 [batch_size, seq_len, d_model]
    for ape_module in self.ape:
        x_0 = ape_module(x_0) # 绝对位置编码只加在最开始输入处
    x = x_0
    if current_blocks is None:
        current_blocks = self.num_blocks
    b_0 = max(0, current_blocks - num_eff) # 计算需要加入误差计算的有效层数对
应的起始层索引
    outputs = []
    if self.loop:
        self.probe.reset() # 每次前向传播前重置捕获的数据，避免混淆不同次前向
传播的注意力矩阵
        for i in range(current_blocks):
            if i == b_0:
                x = x.detach() # 释放前面的计算图
                x.requires_grad_(True)
            x = self.transformer_block(x+x_0)
            if i >= b_0:
                outputs.append(x)

```

```

        self.captured_attention = list(self.probe.captured_data) # 显式 copy
一份数据，避免后续被修改
    else:
        for probe in self.probe:
            probe.reset()
        for i, block in enumerate(self.transformer_block):
            if i == b_0:
                x = x.detach() # 释放前面的计算图
                x.requires_grad_(True)
                x = block(x+x_0)
            if i >= b_0:
                outputs.append(x)
        self.captured_attention = [probe.captured_data for probe in self.
↪probe]
        outputs = torch.stack(outputs, dim=1) # [batch_size, num_eff, seq_len, ↪
↪d_model]
    return outputs

```

1.2.6 Downstream Task: Linear Regression

转换头

```

[64]: class RegressionHead(nn.Module):
    def __init__(self, d_model, d_x, d_y=1, bias=False, init_scale=None):
        super().__init__()
        self.read_in_x = nn.Linear(d_x, d_model, bias=bias)
        self.read_in_y = nn.Linear(d_y, d_model, bias=bias)
        if init_scale is not None:
            nn.init.normal_(self.read_in_x.weight, mean=0.0, std=init_scale)
            nn.init.normal_(self.read_in_y.weight, mean=0.0, std=init_scale)
    def forward(self, x_data, y_data):
        # k 相当于 seq_len 的一半，因为 x 和 y 交织在一起，所以总的 seq_len 是 2k
        # x_data: [batch_size, k, d_x]
        # y_data: [batch_size, k, d_y]
        x_emb = self.read_in_x(x_data) # [batch_size, k, d_model]
        y_emb = self.read_in_y(y_data) # [batch_size, k, d_model]
        # 交织成 (x_1, y_1, x_2, y_2, ..., x_k, y_k)

```

```

        stacked = torch.stack((x_emb, y_emb), dim=2) # [batch_size, k, 2,
↪ d_model]

        Prompt = stacked.flatten(1,2) # [batch_size, 2k, d_model]

        return Prompt

```

损失函数

```

[65]: class PredictionLoss(nn.Module):
    def __init__(self, d_model, d_y=1, loss_type='mse'):
        super().__init__()
        self.read_out = nn.Linear(d_model, d_y)
        if loss_type == 'mse':
            self.loss_fn = nn.MSELoss()
        elif loss_type == 'l1':
            self.loss_fn = nn.L1Loss()

    def forward(self, outputs, y_true, is_eval=False):
        # outputs: [batch_size, num_eff, seq_len, d_model]
        # y_true: [batch_size, seq_len/2, 1]
        if is_eval:
            y_outputs = outputs[:, -1, -2, :] # 只取最后一层的最后一个位置的输出
            作为预测值

            y_pred_final = self.read_out(y_outputs) # [batch_size, d_y]
            return y_pred_final

            y_outputs = outputs[:, :, 0::2, :] # [batch_size, num_eff, seq_len/2,
↪ d_model]

            y_preds = self.read_out(y_outputs) # [batch_size, num_eff, seq_len/2,
↪ d_y]

            loss = self.loss_fn(y_preds, y_true.unsqueeze(1).expand_as(y_preds)) #
↪ [batch_size, num_eff, seq_len/2, d_y]

            return loss

```

拼装

```

[66]: class RegressionSolver(nn.Module):
    def __init__(self, num_blocks, num_heads, d_model, d_x, d_y, max_seq_len,
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
↪ b_rope_or_upe=10000, head_ratio_upe=2,

```

```

        loop=True, loss_type='mse',
        bias=False, init_scale=None):
    super().__init__()
    self.toy_model = ToyModel(num_blocks, num_heads, d_model, max_seq_len,
    ↪ norm_type=norm_type, ffn_type=ffn_type, pe_type=pe_type, loop=loop,
    ↪ b_rope_or_upe=b_rope_or_upe, head_ratio_upe=head_ratio_upe)
    self.head = RegressionHead(d_model, d_x, d_y, bias=bias,
    ↪ init_scale=init_scale)
    self.loss_fn = PredictionLoss(d_model, d_y, loss_type=loss_type)

    def forward(self, x_data, y_data, num_eff, current_blocks=None,
    ↪ is_eval=False):
        Prompt = self.head(x_data, y_data) # [batch_size, seq_len, d_model]
        outputs = self.toy_model(Prompt, num_eff,
    ↪ current_blocks=current_blocks) # [batch_size, num_eff, seq_len, d_model]
        output = self.loss_fn(outputs, y_data, is_eval=is_eval)
        return output

```

1.2.7 Experiment

1.2.8 数据生成

线性回归数据

```

[67]: def linear_data_generator(batch_size, seq_len, d_x, d_y, device='cpu'):
    w = torch.randn(batch_size, d_x, d_y, device=device) # 真实线性函数 w:
    ↪ [batch_size, d_x, d_y]
    x_data = torch.randn(batch_size, seq_len//2, d_x, device=device) # 输入特
    征 x: [batch_size, seq_len//2, d_x]
    y_data = x_data @ w # 计算标签 y = x @ w: [batch_size, seq_len//2, d_y]
    return x_data, y_data

```

1.2.9 数据加载器

```

[ ]: def dataloader(batch_size=64, seq_len=80, d_x=20, d_y=1, device='cpu',
    ↪ data_type='linear', sink_padding=None):
    while True:
        if data_type == 'linear':

```

```

        yield linear_data_generator(batch_size, seq_len, d_x, d_y,
↪device=device)

```

1.2.10 实验模块

```

[69]: class LoopedTransformerExperiment:
    def __init__(self, num_blocks, num_heads=8, d_model=256, lr=1e-4,
↪max_seq_len=4096, d_x=20, d_y=1, seed=None, experiment_name='default_exp',
        norm_type='layernorm', ffn_type='gelu', pe_type='learned_ape',
↪b_rope_or_upe=10000, head_ratio_upe=2,
        optimizer_type='adam', loop=True, loss_type='mse', bias=False,
↪init_scale=None,
        task='regression'):
        # 设备选择: 优先使用 MPS (适用于 Apple Silicon), 其次是 CUDA (适用于
↪NVIDIA GPU), 最后退回 CPU
        if torch.backends.mps.is_available():
            self.device = torch.device("mps")
        elif torch.cuda.is_available():
            self.device = torch.device("cuda")
        else:
            self.device = torch.device("cpu")
        if experiment_name is not None:
            print(f"{experiment_name} initialized. Using device: {self.device}")
        if seed is not None:
            torch.manual_seed(seed)
        # 模型和优化器
        self.task = task
        if task == 'regression':
            self.model = RegressionSolver(num_blocks, num_heads, d_model,
↪d_x=d_x, d_y=d_y, max_seq_len=max_seq_len,
                norm_type=norm_type,
↪ffn_type=ffn_type, pe_type=pe_type, b_rope_or_upe=b_rope_or_upe,
                head_ratio_upe=head_ratio_upe,
↪loop=loop, loss_type=loss_type,
                bias=bias, init_scale=init_scale).
↪to(self.device)

```

```

if optimizer_type == 'adam':
    self.optimizer = torch.optim.Adam(self.model.parameters(), lr=lr)
elif optimizer_type == 'sgd':
    self.optimizer = torch.optim.SGD(self.model.parameters(), lr=lr)
self.d_x = d_x
self.d_y = d_y
self.num_blocks = num_blocks

def train(self, batch_size=64, seq_len=80, epochs=20, data_type='linear',
num_eff=15, scheduled_training=True, print_every=None, timing=False):
    # x_data: [batch_size, k, d_x]
    # y_data: [batch_size, k, d_y]
    self.model.train()
    self.data_loader = dataloader(batch_size, seq_len, self.d_x, self.d_y,
device=self.device, data_type=data_type)
    self.loss_history = []
    num_eff = min(num_eff, self.num_blocks) # 确保 num_eff 不超过总层数
    if timing:
        start_time = time.time()
    for epoch in range(epochs):
        self.optimizer.zero_grad()
        x_data, y_data = next(self.data_loader) # 从数据加载器中获取一个批次
的数据
        if scheduled_training:
            current_blocks = min(num_eff + epoch, self.num_blocks) # 随着训
练的进行, 逐渐增加参与误差计算的有效层数
        else:
            current_blocks = self.num_blocks
        loss = self.model(x_data, y_data, num_eff=num_eff,
current_blocks=current_blocks, is_eval=False) # 前向传播计算损失 [batch_size,
num_eff, seq_len/2, d_y]
        loss = loss.mean() # 对所有维度求平均得到一个标量损失值
        loss.backward() # 反向传播计算梯度
        self.optimizer.step() # 更新模型参数
        self.loss_history.append(loss.item()) # 记录损失值
        if print_every is not None and (epoch + 1) % print_every == 0:

```



```

        print(f"Epoch {epoch+1}/{epochs}, Loss: {loss.item():.4f},  

↪current_blocks: {current_blocks}")
        if timing:
            end_time = time.time()
            print(f"Training completed in {end_time - start_time:.2f} seconds.")

    def evaluate(self, batch_size=64, seq_len=80, data_type='linear',  

↪loss_type='mse'):
        self.model.eval()
        with torch.no_grad():
            # 在评估模式下, 直接使用所有层的输出进行预测 (因为已经 no_grad 了, 所以  

            # 不需要防止梯度爆炸了)
            x_eval, y_eval = next(dataloader(batch_size, seq_len+1, self.d_x,  

↪self.d_y, device=self.device, data_type=data_type))
            y_pred = self.model(x_eval, y_eval, num_eff=self.num_blocks,  

↪current_blocks=self.num_blocks, is_eval=True) # [batch_size, d_y]
            if loss_type == 'mse':
                mse=F.mse_loss(y_pred, y_eval[:, -1, :])
                print(f"MSELoss: {mse:.4f}") # 计算预测值与真实值之间的均方误差
            elif loss_type == 'l1':
                l1=F.l1_loss(y_pred, y_eval[:, -1, :])
                print(f"L1Loss: {l1:.4f}")
        return y_pred

```

1.3 主程序

1.3.1 实验数据观测

绘制 Loss 下降曲线

```

[71]: init_parameters = {
        'num_blocks': 20,
        'num_heads': 8,
        'd_model': 256,
        'lr': 1e-4,
        'max_seq_len': 100,
        'd_x': 20,
        'd_y': 1,

```

```

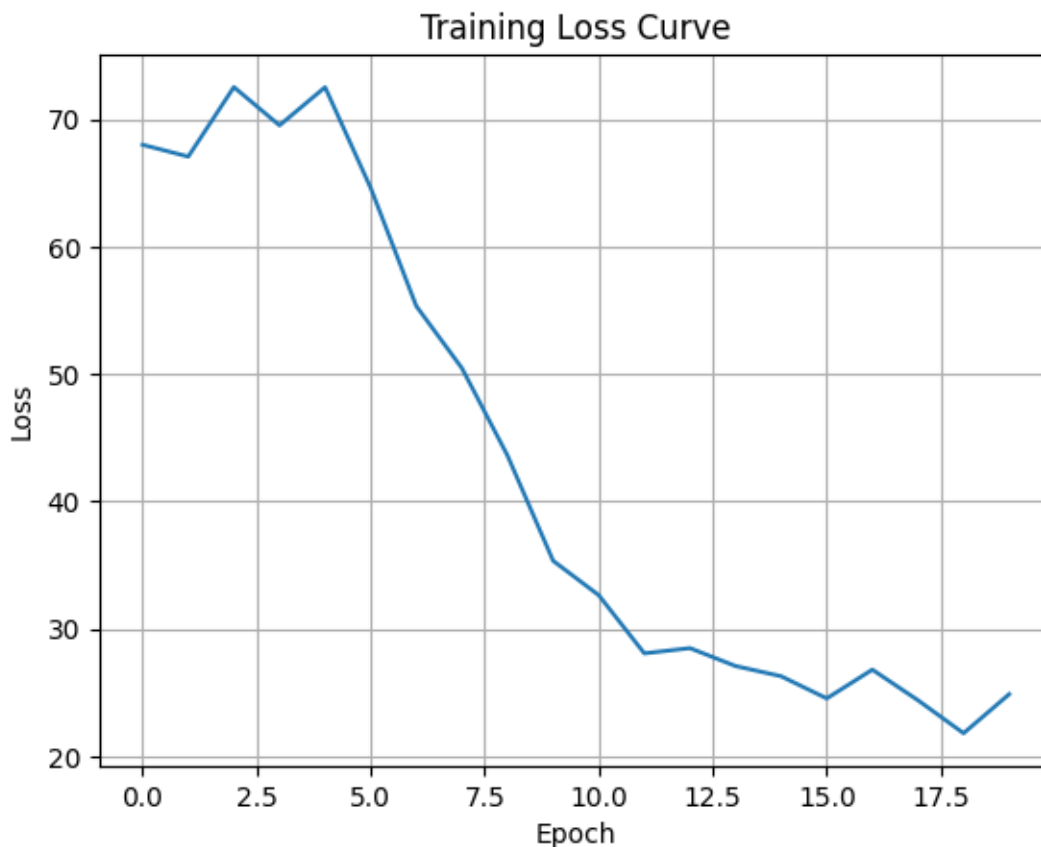
    'seed': 42,
    'experiment_name': 'Experiment',
    'norm_type': 'layernorm',
    'ffn_type': 'gelu',
    'pe_type': ['learned_ape'],
    'optimizer_type': 'adam',
    'loop': True,
    'loss_type': 'mse',
    'bias': False,
    'init_scale': None,
    'task': 'regression'
}
train_parameters = {
    'batch_size': 64,
    'seq_len': 80,
    'epochs': 20,
    'num_eff': 15,
    'data_type': 'linear',
    'scheduled_training': True,
    'print_every': 5,
    'timing': True
}
experiment = LoopedTransformerExperiment(**init_parameters)
experiment.train(**train_parameters)
plt.plot(experiment.loss_history)
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Training Loss Curve')
plt.grid()
plt.show()

```

```

Experiment initialized. Using device: mps
Epoch 5/20, Loss: 72.5179, current_blocks: 19
Epoch 10/20, Loss: 35.3423, current_blocks: 20
Epoch 15/20, Loss: 26.2775, current_blocks: 20
Epoch 20/20, Loss: 24.8684, current_blocks: 20
Training completed in 9.05 seconds.

```



1.3.2 对比实验

Scheduled Training vs Non-Scheduled Training 对比实验

```
[72]: init_parameters['seed'] = 42
init_parameters['experiment_name'] = 'Experiment scheduled'
train_parameters['scheduled_training'] = True
train_parameters['print_every'] = 2
train_parameters['timing'] = True
experiment1 = LoopedTransformerExperiment(**init_parameters)
experiment1.train(**train_parameters)

init_parameters['experiment_name'] = 'Experiment non-scheduled'
train_parameters['scheduled_training'] = False
experiment2 = LoopedTransformerExperiment(**init_parameters)
experiment2.train(**train_parameters)
```

```

# 绘制 Loss 下降曲线
# 在一张图上绘制两条曲线，分别对应 scheduled_training=True 和 False 的情况
# 图例（不同颜色代表不同的训练方式）
plt.plot(experiment1.loss_history, label='Scheduled Training', color='blue')
plt.plot(experiment2.loss_history, label='Non-Scheduled Training', color='red')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Scheduled Training vs Non-Scheduled Training')
plt.grid()
plt.legend()
plt.show()

```

Experiment scheduled initialized. Using device: mps

```

Epoch 2/20, Loss: 67.0742, current_blocks: 16
Epoch 4/20, Loss: 69.5261, current_blocks: 18
Epoch 6/20, Loss: 64.6115, current_blocks: 20
Epoch 8/20, Loss: 50.4657, current_blocks: 20
Epoch 10/20, Loss: 35.3423, current_blocks: 20
Epoch 12/20, Loss: 28.0742, current_blocks: 20
Epoch 14/20, Loss: 27.0792, current_blocks: 20
Epoch 16/20, Loss: 24.5359, current_blocks: 20
Epoch 18/20, Loss: 24.3915, current_blocks: 20
Epoch 20/20, Loss: 24.8684, current_blocks: 20
Training completed in 6.66 seconds.

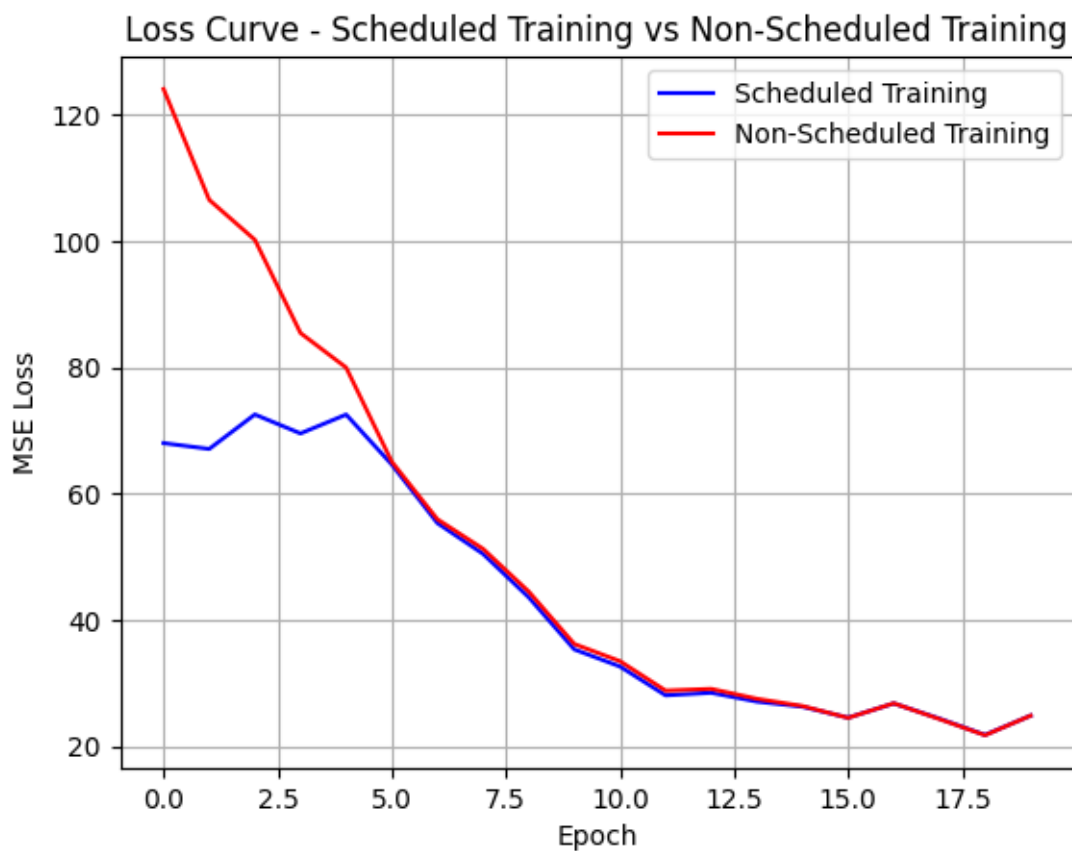
```

Experiment non-scheduled initialized. Using device: mps

```

Epoch 2/20, Loss: 106.4994, current_blocks: 20
Epoch 4/20, Loss: 85.4250, current_blocks: 20
Epoch 6/20, Loss: 65.0549, current_blocks: 20
Epoch 8/20, Loss: 51.2620, current_blocks: 20
Epoch 10/20, Loss: 36.1977, current_blocks: 20
Epoch 12/20, Loss: 28.8228, current_blocks: 20
Epoch 14/20, Loss: 27.5290, current_blocks: 20
Epoch 16/20, Loss: 24.5093, current_blocks: 20
Epoch 18/20, Loss: 24.3212, current_blocks: 20
Epoch 20/20, Loss: 24.8631, current_blocks: 20
Training completed in 6.59 seconds.

```



不同 Positional Embedding 对比实验

```
[85]: init_parameters = {  
    'num_blocks': 20,  
    'num_heads': 8,  
    'd_model': 256,  
    'lr': 1e-4,  
    'max_seq_len': 100,  
    'd_x': 2,  
    'd_y': 1,  
    'seed': 46,  
    'experiment_name': 'learned_ape_exp',  
    'norm_type': 'layernorm',  
    'ffn_type': 'gelu',  
    'pe_type': ['learned_ape'],  
}
```

```

    'loop': True,
    'loss_type': 'mse',
    'bias': False,
    'init_scale': None
}
train_parameters = {
    'batch_size': 64,
    'seq_len': 80,
    'epochs': 100,
    'num_eff': 15,
    'data_type': 'linear',
    'scheduled_training': True,
    'print_every': 5,
    'timing': True
}

```

```

[86]: learning_ape_exp = LoopedTransformerExperiment(**init_parameters)
      learning_ape_exp.train(**train_parameters)

```

```

learned_ape_exp initialized. Using device: mps
Epoch 5/100, Loss: 41.1805, current_blocks: 19
Epoch 10/100, Loss: 18.3069, current_blocks: 20
Epoch 15/100, Loss: 7.0673, current_blocks: 20
Epoch 20/100, Loss: 5.1770, current_blocks: 20
Epoch 25/100, Loss: 5.9387, current_blocks: 20
Epoch 30/100, Loss: 4.4391, current_blocks: 20
Epoch 35/100, Loss: 3.7256, current_blocks: 20
Epoch 40/100, Loss: 3.1097, current_blocks: 20
Epoch 45/100, Loss: 3.2062, current_blocks: 20
Epoch 50/100, Loss: 2.9484, current_blocks: 20
Epoch 55/100, Loss: 2.6715, current_blocks: 20
Epoch 60/100, Loss: 2.9714, current_blocks: 20
Epoch 65/100, Loss: 2.3141, current_blocks: 20
Epoch 70/100, Loss: 2.6434, current_blocks: 20
Epoch 75/100, Loss: 2.3794, current_blocks: 20
Epoch 80/100, Loss: 2.3924, current_blocks: 20
Epoch 85/100, Loss: 2.8761, current_blocks: 20

```

Epoch 90/100, Loss: 3.1192, current_blocks: 20
Epoch 95/100, Loss: 3.2056, current_blocks: 20
Epoch 100/100, Loss: 2.6971, current_blocks: 20
Training completed in 34.92 seconds.

```
[87]: init_parameters['experiment_name'] = 'ape_exp'
      init_parameters['pe_type'] = 'ape'
      ape_exp = LoopedTransformerExperiment(**init_parameters)
      ape_exp.train(**train_parameters)
```

ape_exp initialized. Using device: mps
Epoch 5/100, Loss: 10.5527, current_blocks: 19
Epoch 10/100, Loss: 9.5900, current_blocks: 20
Epoch 15/100, Loss: 6.6546, current_blocks: 20
Epoch 20/100, Loss: 5.5995, current_blocks: 20
Epoch 25/100, Loss: 4.4722, current_blocks: 20
Epoch 30/100, Loss: 3.7418, current_blocks: 20
Epoch 35/100, Loss: 3.0496, current_blocks: 20
Epoch 40/100, Loss: 2.7531, current_blocks: 20
Epoch 45/100, Loss: 2.7307, current_blocks: 20
Epoch 50/100, Loss: 2.3372, current_blocks: 20
Epoch 55/100, Loss: 2.0070, current_blocks: 20
Epoch 60/100, Loss: 2.4516, current_blocks: 20
Epoch 65/100, Loss: 1.9479, current_blocks: 20
Epoch 70/100, Loss: 2.1836, current_blocks: 20
Epoch 75/100, Loss: 1.9636, current_blocks: 20
Epoch 80/100, Loss: 2.0577, current_blocks: 20
Epoch 85/100, Loss: 2.4641, current_blocks: 20
Epoch 90/100, Loss: 2.7406, current_blocks: 20
Epoch 95/100, Loss: 2.6699, current_blocks: 20
Epoch 100/100, Loss: 2.2478, current_blocks: 20
Training completed in 33.32 seconds.

```
[88]: init_parameters['experiment_name'] = 'alibi_exp'
      init_parameters['pe_type'] = 'alibi'
      alibi_exp = LoopedTransformerExperiment(**init_parameters)
      alibi_exp.train(**train_parameters)
```

```
alibi_exp initialized. Using device: mps
Epoch 5/100, Loss: 7.9440, current_blocks: 19
Epoch 10/100, Loss: 5.0251, current_blocks: 20
Epoch 15/100, Loss: 4.0158, current_blocks: 20
Epoch 20/100, Loss: 3.2134, current_blocks: 20
Epoch 25/100, Loss: 3.3868, current_blocks: 20
Epoch 30/100, Loss: 2.5129, current_blocks: 20
Epoch 35/100, Loss: 2.3932, current_blocks: 20
Epoch 40/100, Loss: 2.3417, current_blocks: 20
Epoch 45/100, Loss: 2.4873, current_blocks: 20
Epoch 50/100, Loss: 1.9661, current_blocks: 20
Epoch 55/100, Loss: 1.7287, current_blocks: 20
Epoch 60/100, Loss: 2.0730, current_blocks: 20
Epoch 65/100, Loss: 1.7679, current_blocks: 20
Epoch 70/100, Loss: 2.1053, current_blocks: 20
Epoch 75/100, Loss: 1.9362, current_blocks: 20
Epoch 80/100, Loss: 1.7442, current_blocks: 20
Epoch 85/100, Loss: 2.1672, current_blocks: 20
Epoch 90/100, Loss: 2.5389, current_blocks: 20
Epoch 95/100, Loss: 2.2342, current_blocks: 20
Epoch 100/100, Loss: 2.1209, current_blocks: 20
Training completed in 34.19 seconds.
```

```
[89]: init_parameters['experiment_name'] = 'rope_exp'
      init_parameters['pe_type'] = 'rope'
      init_parameters['b_rope_or_upe'] = 10000
      rope_exp = LoopedTransformerExperiment(**init_parameters)
      rope_exp.train(**train_parameters)
```

```
rope_exp initialized. Using device: mps
Epoch 5/100, Loss: 8.5121, current_blocks: 19
Epoch 10/100, Loss: 5.3167, current_blocks: 20
Epoch 15/100, Loss: 3.9336, current_blocks: 20
Epoch 20/100, Loss: 3.2812, current_blocks: 20
Epoch 25/100, Loss: 3.6321, current_blocks: 20
Epoch 30/100, Loss: 2.6115, current_blocks: 20
Epoch 35/100, Loss: 2.4616, current_blocks: 20
```



```
Epoch 40/100, Loss: 2.4041, current_blocks: 20
Epoch 45/100, Loss: 2.5846, current_blocks: 20
Epoch 50/100, Loss: 2.0036, current_blocks: 20
Epoch 55/100, Loss: 1.7312, current_blocks: 20
Epoch 60/100, Loss: 2.1332, current_blocks: 20
Epoch 65/100, Loss: 1.7742, current_blocks: 20
Epoch 70/100, Loss: 2.0761, current_blocks: 20
Epoch 75/100, Loss: 1.9971, current_blocks: 20
Epoch 80/100, Loss: 1.7743, current_blocks: 20
Epoch 85/100, Loss: 2.2035, current_blocks: 20
Epoch 90/100, Loss: 2.6322, current_blocks: 20
Epoch 95/100, Loss: 2.3430, current_blocks: 20
Epoch 100/100, Loss: 2.2058, current_blocks: 20
Training completed in 46.83 seconds.
```

```
[90]: init_parameters['experiment_name'] = 'ms_upe_exp'
      init_parameters['pe_type'] = 'ms_upe'
      init_parameters['b_rope_or_upe'] = 10000
      init_parameters['head_ratio_upe'] = 2
      ms_upe_exp = LoopedTransformerExperiment(**init_parameters)
      ms_upe_exp.train(**train_parameters)
```

```
ms_upe_exp initialized. Using device: mps
Epoch 5/100, Loss: 8.0830, current_blocks: 19
Epoch 10/100, Loss: 5.2086, current_blocks: 20
Epoch 15/100, Loss: 4.1812, current_blocks: 20
Epoch 20/100, Loss: 2.9801, current_blocks: 20
Epoch 25/100, Loss: 3.2729, current_blocks: 20
Epoch 30/100, Loss: 2.6667, current_blocks: 20
Epoch 35/100, Loss: 2.5216, current_blocks: 20
Epoch 40/100, Loss: 2.3067, current_blocks: 20
Epoch 45/100, Loss: 2.4673, current_blocks: 20
Epoch 50/100, Loss: 2.0190, current_blocks: 20
Epoch 55/100, Loss: 1.7739, current_blocks: 20
Epoch 60/100, Loss: 2.1786, current_blocks: 20
Epoch 65/100, Loss: 1.8399, current_blocks: 20
Epoch 70/100, Loss: 2.0640, current_blocks: 20
```

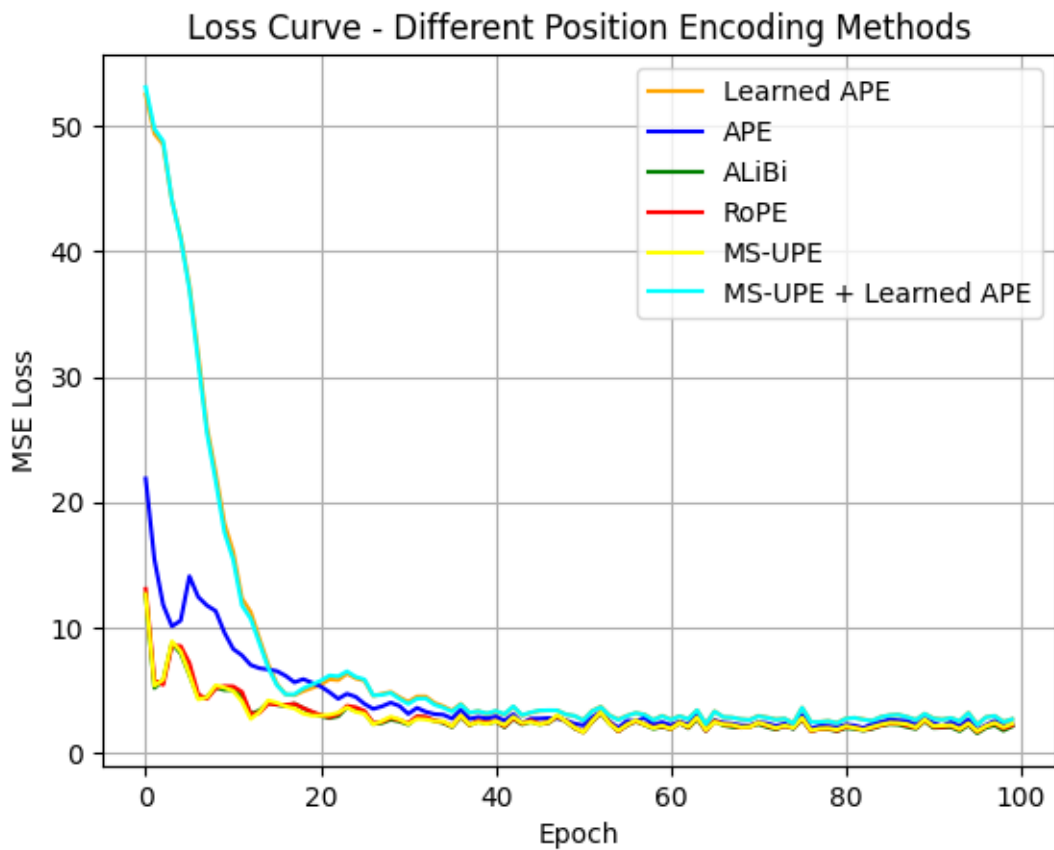
```
Epoch 75/100, Loss: 2.0052, current_blocks: 20
Epoch 80/100, Loss: 1.8559, current_blocks: 20
Epoch 85/100, Loss: 2.2586, current_blocks: 20
Epoch 90/100, Loss: 2.6707, current_blocks: 20
Epoch 95/100, Loss: 2.3728, current_blocks: 20
Epoch 100/100, Loss: 2.3211, current_blocks: 20
Training completed in 39.21 seconds.
```

```
[91]: init_parameters['experiment_name'] = 'ms_upe&learned_ape_exp'
init_parameters['pe_type'] = ['ms_upe', 'learned_ape']
init_parameters['b_rope_or_upe'] = 10000
init_parameters['head_ratio_upe'] = 2
ms_upe_learned_ape_exp = LoopedTransformerExperiment(**init_parameters)
ms_upe_learned_ape_exp.train(**train_parameters)
```

```
ms_upe&learned_ape_exp initialized. Using device: mps
```

```
Epoch 5/100, Loss: 41.0124, current_blocks: 19
Epoch 10/100, Loss: 17.5966, current_blocks: 20
Epoch 15/100, Loss: 6.7853, current_blocks: 20
Epoch 20/100, Loss: 5.4592, current_blocks: 20
Epoch 25/100, Loss: 6.0685, current_blocks: 20
Epoch 30/100, Loss: 4.3091, current_blocks: 20
Epoch 35/100, Loss: 3.6125, current_blocks: 20
Epoch 40/100, Loss: 3.1510, current_blocks: 20
Epoch 45/100, Loss: 3.2357, current_blocks: 20
Epoch 50/100, Loss: 2.9586, current_blocks: 20
Epoch 55/100, Loss: 2.6367, current_blocks: 20
Epoch 60/100, Loss: 2.9363, current_blocks: 20
Epoch 65/100, Loss: 2.3051, current_blocks: 20
Epoch 70/100, Loss: 2.6297, current_blocks: 20
Epoch 75/100, Loss: 2.3529, current_blocks: 20
Epoch 80/100, Loss: 2.3792, current_blocks: 20
Epoch 85/100, Loss: 2.8603, current_blocks: 20
Epoch 90/100, Loss: 3.1174, current_blocks: 20
Epoch 95/100, Loss: 3.1854, current_blocks: 20
Epoch 100/100, Loss: 2.6862, current_blocks: 20
Training completed in 41.03 seconds.
```

```
[92]: plt.plot(learning_ape_exp.loss_history, label='Learned APE', color='orange')
plt.plot(ape_exp.loss_history, label='APE', color='blue')
plt.plot(alibi_exp.loss_history, label='ALiBi', color='green')
plt.plot(rope_exp.loss_history, label='RoPE', color='red')
plt.plot(ms_upe_exp.loss_history, label='MS-UPE', color='yellow')
plt.plot(ms_upe_learned_ape_exp.loss_history, label='MS-UPE + Learned APE',
        color='cyan')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Different Position Encoding Methods')
plt.grid()
plt.legend()
plt.show()
```



```
[93]: from operator import sub
plt.plot(list(map(sub, alibi_exp.loss_history, alibi_exp.loss_history)),
        label='ALiBi-ALiBi', color='yellow')
plt.plot(list(map(sub, rope_exp.loss_history, alibi_exp.loss_history)),
        label='RoPE-ALiBi', color='blue')
plt.plot(list(map(sub, ms_upe_exp.loss_history, alibi_exp.loss_history)),
        label='MS-UPE-ALiBi', color='red')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.title('Loss Curve - Relative Loss Compared to ALiBi')
plt.grid()
plt.legend()
plt.show()
```

